

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398580
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Zhao; Yong et al.

Mounting assembly for door lock

Abstract

A door lock that includes a latch adapted to be moved manually relative to a mounting bracket and lock body to attach the body to the mounting bracket. The latch may be movably mounted to the body, and arranged to capture a portion of the mounting bracket between the latch and the body. A detent may be used to hold the latch in the closed position relative to the body, and a spring may be adapted to bias the latch away from the body.

Inventors: Zhao; Yong (Medford, MA), Marvin; Heath Allen (Woburn, MA), Futran; Chaim Carmiel (Cambridge, MA)

Applicant: SimpliSafe, Inc. (Boston, MA)

Family ID: 1000008778302

Assignee: SimpliSafe, Inc. (Boston, MA)

Appl. No.: 18/401884

Filed: January 02, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240151064 A1	May. 09, 2024

Related U.S. Application Data

continuation parent-doc US 16539214 20190813 US 11933092 child-doc US 18401884

Publication Classification

Int. Cl.: E05B9/08 (20060101)

U.S. Cl.:

CPC **E05B9/08** (20130101); E05Y2900/132 (20130101)

Field of Classification Search

CPC: H04L (5/0055); H04L (5/0053); H04L (5/0057); H04L (5/001); H04L (5/003); H04L (5/0091); H04L (5/0048); H04L (1/1812); H04L (1/08); H04L (1/1671); H04L (1/1861); H04L (1/20); H04W (72/044); H04W (72/20); H04W (72/23); H04W (72/0453); H04W (72/046); H04W (72/262); H04W (72/0626); H04W (72/0632); H04W (72/0639); H04W (72/06952); H04W (72/088); H04W (72/0457); H04W (72/21); H04W (72/231)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3397905	12/1967	Banse	N/A	N/A
4156541	12/1978	Babb, Jr. et al.	N/A	N/A
4845490	12/1988	Ward et al.	N/A	N/A
5369913	12/1993	Brickner	N/A	N/A
5421178	12/1994	Hamel et al.	N/A	N/A
5460863	12/1994	Kessel et al.	N/A	N/A
5473236	12/1994	Frolov	N/A	N/A
5474348	12/1994	Palmer et al.	N/A	N/A
5628216	12/1996	Qureshi et al.	N/A	N/A
5646605	12/1996	Leonaggeo et al.	N/A	N/A
5712626	12/1997	Andreou et al.	N/A	N/A
5782118	12/1997	Chamberlain et al.	N/A	N/A
5857365	12/1998	Armstrong	N/A	N/A
5931430	12/1998	Palmer	N/A	N/A
6076870	12/1999	Frolov	N/A	N/A
6330817	12/2000	Frolov	N/A	N/A
6354121	12/2001	Frolov	N/A	N/A
6442986	12/2001	Russell et al.	N/A	N/A
D552964	12/2006	Bogdanov et al.	N/A	N/A
7497103	12/2008	Misner	70/120	E05B 63/14
9685017	12/2016	Johnson	N/A	N/A
11933092	12/2023	Zhao et al.	N/A	N/A
2001/0025517	12/2000	Bates et al.	N/A	N/A
2002/0095958	12/2001	Bates et al.	N/A	N/A
2003/0179074	12/2002	Palomaki et al.	N/A	N/A
2004/0177663	12/2003	Walsh, III et al.	N/A	N/A
2005/0050928	12/2004	Frolov et al.	N/A	N/A
2005/0167995	12/2004	Wallis et al.	N/A	N/A
2005/0179267	12/2004	Cote et al.	N/A	N/A
2006/0101878	12/2005	Dickhans et al.	N/A	N/A
2007/0157684	12/2006	Bogdanov et al.	N/A	N/A

2007/0204663	12/2006	Lee	N/A	N/A
2007/0209413	12/2006	Dobbs	N/A	N/A
2008/0142593	12/2007	Walsh et al.	N/A	N/A
2008/0169655	12/2007	Gomez Gonzalez	N/A	N/A
2008/0236213	12/2007	Blanch	N/A	N/A
2009/0027197	12/2008	Frolov	N/A	N/A
2009/0277232	12/2008	Walsh, III	N/A	N/A
2009/0282877	12/2008	Laverty et al.	N/A	N/A
2010/0089109	12/2009	Bliding et al.	N/A	N/A
2010/0257906	12/2009	Sorensen et al.	N/A	N/A
2010/0294008	12/2009	Bogdanov et al.	N/A	N/A
2011/0036131	12/2010	Chiou et al.	N/A	N/A
2011/0067464	12/2010	Chiou et al.	N/A	N/A
2011/0203331	12/2010	Picard et al.	N/A	N/A
2012/0061974	12/2011	Laverty et al.	N/A	N/A
2012/0212001	12/2011	Benitez et al.	N/A	N/A
2012/0324967	12/2011	Goren et al.	N/A	N/A
2013/0031940	12/2012	Romero	N/A	N/A
2013/0039715	12/2012	Daykin et al.	N/A	N/A
2013/0061644	12/2012	Trent et al.	N/A	N/A
2013/0192316	12/2012	McKibben et al.	N/A	N/A
2013/0318519	12/2012	Coolidge	N/A	N/A
2013/0335222	12/2012	Comerford et al.	N/A	N/A
2014/0109633	12/2013	Romero	N/A	N/A
2014/0109634	12/2013	Almomani et al.	N/A	N/A
2014/0113563	12/2013	Almomani et al.	N/A	N/A
2014/0118107	12/2013	Almomani	N/A	N/A
2014/0157842	12/2013	Almomani et al.	N/A	N/A
2014/0159392	12/2013	Tsai et al.	N/A	N/A
2014/0165673	12/2013	Tyner et al.	N/A	N/A
2014/0250956	12/2013	Chong	N/A	N/A
2014/0260448	12/2013	Beck et al.	N/A	N/A
2014/0260449	12/2013	Uyeda et al.	N/A	N/A
2014/0260451	12/2013	Quach et al.	N/A	N/A
2014/0265359	12/2013	Cheng et al.	N/A	N/A
2014/0292001	12/2013	Nunez et al.	N/A	N/A
2015/0184425	12/2014	Ellis et al.	N/A	N/A
2015/0240528	12/2014	Mani et al.	N/A	N/A
2015/0240529	12/2014	Dore Vasudevan et al.	N/A	N/A
2015/0257345	12/2014	Carpenter et al.	N/A	N/A
2015/0281228	12/2014	Rettig et al.	N/A	N/A
2015/0361704	12/2014	Larsen et al.	N/A	N/A
2016/0040452	12/2015	Ku	N/A	N/A
2016/0060904	12/2015	Dore Vasudevan et al.	N/A	N/A
2016/0115720	12/2015	Wong et al.	N/A	N/A
2016/0145898	12/2015	Ohl et al.	N/A	N/A
2016/0168888	12/2015	Farias et al.	N/A	N/A

2016/0186463	12/2015	Roatis	70/266	E05C 3/12
2016/0253855	12/2015	Lowder	N/A	N/A
2016/0340930	12/2015	Schaeffer	N/A	E05B 15/02
2017/0191287	12/2016	Mittleman et al.	N/A	N/A
2017/0204637	12/2016	Matyko	N/A	N/A
2017/0342742	12/2016	Snodgrass	N/A	N/A
2017/0342750	12/2016	Hiramine	N/A	N/A
2018/0061163	12/2017	Ku	N/A	N/A
2018/0108228	12/2017	Lovejoy	N/A	N/A
2018/0320412	12/2017	Matosian	N/A	N/A
2018/0320413	12/2017	Wong	N/A	N/A
2018/0340352	12/2017	Compton	N/A	N/A
2019/0019365	12/2018	Lee	N/A	N/A
2019/0211581	12/2018	Roup et al.	N/A	N/A
2019/0257116	12/2018	Coster et al.	N/A	N/A
2019/0309539	12/2018	Mukundala et al.	N/A	N/A
2019/0383057	12/2018	Larson	N/A	N/A
2020/0002973	12/2019	Hill	N/A	N/A
2020/0011088	12/2019	Harris	N/A	N/A
2020/0011090	12/2019	Ku et al.	N/A	N/A
2020/0032551	12/2019	Holmes et al.	N/A	N/A
2020/0071955	12/2019	Binek et al.	N/A	N/A
2020/0123808	12/2019	Lovejoy et al.	N/A	N/A
2020/0149317	12/2019	Seacat et al.	N/A	N/A
2020/0149320	12/2019	Pfunder	N/A	N/A
2020/0157851	12/2019	Fodstad et al.	N/A	N/A
2020/0248484	12/2019	Arlinghaus et al.	N/A	N/A
2020/0263449	12/2019	Rai et al.	N/A	N/A
2020/0291678	12/2019	Eickhoff et al.	N/A	N/A
2021/0040775	12/2020	Kusanale et al.	N/A	N/A
2021/0047861	12/2020	Zhao et al.	N/A	N/A
2021/0054657	12/2020	Luebeck et al.	N/A	N/A
2021/0079691	12/2020	Sundkvist et al.	N/A	N/A
2021/0148136	12/2020	Kim et al.	N/A	N/A
2021/0174620	12/2020	Kusanale et al.	N/A	N/A
2021/0214970	12/2020	Yang et al.	N/A	N/A
2021/0348422	12/2020	Kumar et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2011224106	12/2011	AU	N/A
104213759	12/2013	CN	N/A
105545092	12/2015	CN	N/A
105840003	12/2015	CN	N/A
206110869	12/2016	CN	N/A
206903396	12/2017	CN	N/A
107795195	12/2017	CN	N/A

207620532	12/2017	CN	N/A
108643715	12/2017	CN	N/A
207988685	12/2017	CN	N/A
110242112	12/2018	CN	N/A
110388138	12/2018	CN	N/A
112343426	12/2020	CN	N/A
112396732	12/2020	CN	N/A
213774876	12/2020	CN	N/A
113250536	12/2020	CN	N/A
41 02 892	12/1991	DE	N/A
10 2006 001 478	12/2006	DE	N/A
10 2011 081 789	12/2011	DE	N/A
20 2013 103 126	12/2012	DE	N/A
0 505 887	12/1991	EP	N/A
2 136 019	12/2008	EP	N/A
2 208 840	12/2009	EP	N/A
2 746 497	12/2013	EP	N/A
3 599 329	12/2019	EP	N/A
2 568 730	12/2018	GB	N/A
2010-112002	12/2009	JP	N/A
2011-017242	12/2010	JP	N/A
2015-202767	12/2014	JP	N/A
10-2005-0074820	12/2004	KR	N/A
10-2006-0039712	12/2005	KR	N/A
20-0468708	12/2012	KR	N/A
20-0484625	12/2016	KR	N/A
10-2018-0083157	12/2017	KR	N/A
20-0486963	12/2017	KR	N/A
20-2020-0001831	12/2019	KR	N/A
WO 01/12931	12/2000	WO	N/A
WO 2004/088069	12/2003	WO	N/A
WO 2007/020623	12/2006	WO	N/A
WO 2007/065200	12/2006	WO	N/A
WO 2007/104499	12/2006	WO	N/A
WO 2012/015135	12/2011	WO	N/A
WO 2012/116037	12/2011	WO	N/A
WO-2017142472	12/2016	WO	E05B 47/02
WO 2019/018897	12/2018	WO	N/A
WO 2019/044962	12/2018	WO	N/A
WO 2019/071306	12/2018	WO	N/A
WO 2020/123573	12/2019	WO	N/A
WO 2020/123575	12/2019	WO	N/A
WO 2021/053435	12/2020	WO	N/A
WO 2021/092660	12/2020	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Application No. PCT/US2020/045880, mailed Dec. 8, 2020. cited by applicant

International Preliminary Report on Patentability for International Application No.

PCT/US2020/045880, mailed Feb. 24, 2022. cited by applicant
PCT/US2020/045880, Dec. 8, 2020, International Search Report and Written Opinion. cited by applicant
PCT/US2020/045880, Feb. 24, 2022, International Preliminary Report on Patentability. cited by applicant

Primary Examiner: Fulton; Kristina R

Assistant Examiner: Neubauer; Thomas L

Attorney, Agent or Firm: Wolf, Greenfield & Sacks, P.C.

Background/Summary

RELATED APPLICATIONS (1) This Application is a Continuation of U.S. application Ser. No. 16/539,214, filed Aug. 13, 2019, entitled “MOUNTING ASSEMBLY FOR DOOR LOCK”.

BACKGROUND

1. Field of Invention

(1) This invention relates to a door lock or other lock control mechanism that is operable to lock and unlock a door or similar structure.

2. Related Art

(2) Electronic door lock actuators, including so-called smart locks that are used to actuate existing door lock mechanisms, are known, e.g., as described in US Patent Application Publication US20170037937. Such door lock arrangements can allow a user to both operate the door lock manually, e.g., by operating a thumb turn, and electronically, e.g., by interacting with the door lock via an electronic device such as a smartphone.

SUMMARY OF INVENTION

(3) In one aspect of the invention, a door lock includes a body adapted to be mounted to a door or other structure that has a component which can be opened and closed, such as a window. Typically, the body is mounted to a movable part of the door, but may be mountable to a door jamb or other stationary element of the door. The body may be mounted to a door or other structure by a mounting bracket adapted to be mounted to the door. For example, the mounting bracket may be screwed or otherwise attached to a door, and the body then attached to the mounting bracket. A latch adapted to be moved manually relative to the mounting bracket and the body to attach the body to the mounting bracket, e.g., allowing a user to attach the body to (and/or remove it from) the mounting bracket by hand and without tools. The rear side of the body may be adapted for positioning adjacent a surface (e.g., in contact with) of a door to which the door lock is mounted. A lock drive may be movably mounted to the body and adapted to move a lock mechanism between locked and unlocked positions. The lock mechanism may include a slidable lock bolt, movable latch or other lock element that, when in a locked position, can prevent or otherwise resist movement of the door from a closed position to an open position. The lock drive may be coupled to the lock mechanism to actuate the lock mechanism based on movement of the lock drive. For example, the lock drive may be coupled to the lock mechanism by a tailpiece or other coupling element so that rotation of the lock drive moves the tailpiece and thus the lock mechanism between lock and unlock positions. A thumb turn may be movably mounted to the body and adapted for manual movement between at least two positions, such as open and closed positions. The thumb turn may be coupled to the lock drive such that movement of the thumb turn moves the lock drive, e.g., so that a user can move the thumb turn to move the lock mechanism between lock and unlock

positions. While the lock drive can be moved manually via the thumb turn, the lock drive can be moved by a motorized or other automated drive train arrangement.

(4) In some embodiments, the latch use to attach the body to the mounting bracket may be movably mounted to the body, e.g., the latch may be movable relative to the body in a plane of a door surface to which the mounting bracket is mounted. The latch may be movable relative to the body between open and closed positions, wherein in the open position a portion of the latch is extended laterally from an outer periphery of the body, and in the closed position the latch is positioned within the outer periphery of the body. In some cases, the latch is movable relative to the body to capture a portion of the mounting bracket between the latch and the body. As an example, the mounting bracket may include a plate arranged in a first plane, and a tab attached to the plate and positioned above the first plane. The latch may have a catch to engage the tab to attach the body to the mounting bracket, e.g., by capturing the tab between the latch and the body. The mounting bracket may include a lock mechanism engaging portion that is positioned below the first plane and adapted to secure the mounting bracket to a door to which the door lock is mounted, e.g., by using threaded fasteners that secure the lock mechanism engaging portion to a lock mechanism mounted to a door.

(5) In some embodiments, first and second latches may be provided that are movable toward each other to attach the body to the mounting bracket. For example, the first and second latches may be mounted on opposite sides of the body. The first and second latches may be arranged to be movable toward each other using a thumb and finger of one hand, e.g., by a user squeezing the two latches toward each other to engage the latches with the mounting bracket. In the open position, a portion of the first and second latches may be extended laterally from an outer periphery of the body, e.g., so a user can grip the latches. In the closed position, the first and second latches may be positioned within the outer periphery of the body, e.g., the user may press the latches into a cavity of the body to the outer portions of the latches are flush with or recessed into the body.

(6) A detent may be provided to hold the latch in the closed position (and/or the open position) relative to the body. For example, the detent may include a pin on the latch and a groove on the body, and be arranged such that the pin and groove can engage each other with the latch in the closed position. The latch may be slidably movable relative to the body, and a spring may be adapted to bias the latch away from the body, e.g., the latch may be mounted to a rear side of the body and the spring may bias the latch away from the rear side of the body. This may help position the latch away from a portion of the body so that a tab or other part of the mounting bracket can be received between the latch and body. The latch may be guided in its movement relative to the body, e.g., the body may have a rail or groove to guide movement of the latch relative to the body, and that latch may have a groove or rail to cooperate with the rail or groove of the body. The latch may have an elongated slot to receive a fastener to attach the latch to the body with the elongated slot extending in a same direction as the groove.

(7) Other advantages and novel features of the invention will become apparent from the following detailed description of various non-limiting embodiments when considered in conjunction with the accompanying figures and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the invention are described with reference to the following drawings in which numerals reference like elements, and wherein:

(2) FIG. 1 is a front perspective view of a door lock in an illustrative embodiment;

(3) FIG. 2 is a front perspective view of the FIG. 1 door lock with the thumb turn removed;

(4) FIG. 3 is a rear perspective view of the thumb turn of the FIG. 1 embodiment;

- (5) FIG. 4 is a rear perspective view of the FIG. 1 embodiment;
- (6) FIG. 5 is a rear perspective view of the FIG. 1 embodiment with latches disengaged from a mounting bracket;
- (7) FIG. 6 is a rear view of another door lock embodiment having slidably movable latches;
- (8) FIG. 7 is a rear perspective view of the FIG. 6 embodiment;
- (9) FIG. 8 is front view of the FIG. 6 embodiment;
- (10) FIG. 9 is a rear perspective view of the FIG. 6 embodiment with latches in an open position;
- (11) FIG. 10 is a front view of the FIG. 6 embodiment with latches in an open position;
- (12) FIG. 11 shows a front right perspective view of the latches and mounting bracket of the FIG. 6 embodiment;
- (13) FIG. 12 shows a front right perspective view of the latches of the FIG. 6 embodiment;
- (14) FIG. 13 is a front view of the mounting bracket of the FIG. 6 embodiment;
- (15) FIG. 14 is a rear view of the mounting bracket of the FIG. 6 embodiment; and
- (16) FIG. 15 is a front perspective view of the mounting bracket of the FIG. 6 embodiment.

DETAILED DESCRIPTION

(17) Aspects of the invention are described below by way of one or more illustrative embodiments. It should be understood that the illustrative embodiments described are not intended to limit the aspects of the invention, but rather to help show how one or more aspects of the invention may be implemented in a particular example. Also, aspects of the invention may be implemented alone and/or in combination with other aspects of the invention.

(18) FIG. 1 shows a perspective view of an illustrative door lock 1 that incorporates one or more aspects of the invention and can be used to move an existing door lock mechanism, such as a sliding bolt or latch, between lock and unlock positions. Embodiments below are described in connection with a door lock mechanism that includes a dead bolt-type lock mechanism in which a bolt element is extended from/retracted into a door structure to lock/unlock the door. However, the door lock 1 may be used with other types of lock mechanisms. In this embodiment, the door lock 1 has a body 10 including a base 17 and a cover plate 16. As discussed in more detail below, the body 10 is mounted to a door, window or other surface to which the door lock 1 is to be attached via a mounting bracket 11. The mounting bracket 11 is initially attached to a door or other surface, and thereafter the body 10 may be attached to the mounting bracket 11 to assemble the door lock 1 on the door. For example, the mounting bracket 11 may have one or more mounting pins 14 that are extended into an opening in a door from an inner side of the door and are secured to a lock cylinder assembly (not shown) on an outer side of the door by screws that engage with the mounting pins 14. Such lock cylinder assemblies are well known in the art and allow a user, for example, to operate the door lock mechanism using a key at the outer side of the door. In other arrangements, the mounting bracket 11 may be mounted over an existing thumb turn of a lock installed on a door. As an example, screws that secure the existing thumb turn of the installed lock may be used to secure the mounting bracket 11 to the door. With the mounting bracket 11 secured to the door or other structure, the base 17 and cover plate 16 of the body 10 may be positioned over and attached to the mounting bracket 11. In this embodiment, the base 17 includes latches 15 that are pivotally mounted to the base 17 and have levers that can be swung outwardly away from the base 17 to allow the base 17 to be positioned over the mounting bracket 11. Thereafter, the latches 15 may be pivoted inwardly so that the latches 15 engage corresponding portions of the mounting bracket 11, thereby attaching the base 17 to the mounting bracket 11. This arrangement can allow for relatively easy and tool-free engagement of the base 17 with the mounting bracket 11.

(19) As is common with many fully manual door locks, the door lock 1 includes a thumb turn 13 that allows a user to manually rotate or otherwise actuate the lock mechanism to move a bolt, latch or other lock element between lock and unlock positions. In this embodiment, the thumb turn 13 may be coupled with a lock tailpiece 22 (either pre-existing or provided with the door lock 1) so that the thumb turn 13 can rotate the tailpiece 22, and thereby move the lock mechanism between

lock and unlock positions. Use of a lock tailpiece **22** with the door lock **1** will typically be done when a thumb turn of an existing lock is removed and the door lock **1** is mounted in its place. Where the door lock **1** is mounted over an existing lock thumb turn, the thumb turn **13** of the door lock **1** may be coupled with the existing lock thumb turn, e.g., by providing the door lock **1** with a coupling device that fits over and has a recess to receive the existing thumb turn, as discussed more below.

(20) FIG. **2** shows a view of the door lock **1** with the thumb turn **13** removed. Normally, the thumb turn **13** is mounted to the cover plate **16**, but the thumb turn **13** is shown removed in FIG. **2** to illustrate that the range of motion of the thumb turn **13** may be limited. For example, the cover plate **16** may have a recess in which the thumb turn **13** is received when engaged to the cover plate **16**. An arcuate slot **161** may be included in the recess and engage with a portion of the thumb turn **13** so that the thumb turn **13** can only be rotated relative to the cover plate **16** to an extent permitted by the slot **161**. For example, FIG. **3** shows a rear perspective view of the thumb turn **13** and illustrates a pin **131** that is received in the slot **161**. When the thumb turn **13** is mounted to the cover plate **16**, the pin **131** moves in the slot **161** as the thumb turn **13** is rotated to actuate the lock mechanism between lock and unlock positions. However, the pin **131** and slot **161** limit the range of motion of the thumb turn **13** relative to the cover plate **16**, e.g., in this case to about 180 degrees rotation. Such limitation is not required, however, and the thumb turn **13** need not be limited in its movement relative to the body **10**. Alternately, the thumb turn **13** may be limited to a range of motion that is less than 180 degrees, e.g., 90 degrees or less.

(21) FIG. **4** shows a rear perspective view of the door lock **1**, and illustrates a tailpiece receiver **23** that is coupled to the thumb turn **13** and may receive the tailpiece **22** shown in FIG. **1**. In this embodiment, the tailpiece receiver **23** engages with a tailpiece coupling **133** at the rear of the thumb turn **13**, as can be seen in FIG. **3**. This couples the thumb turn **13** and the tailpiece receiver **23** so that rotation of the thumb turn **13** rotates the tailpiece receiver **23**. In this embodiment, the tailpiece coupling **133** includes a hexagonal opening that receives a hexagonal portion of the tailpiece receiver **23** to rotationally fix the tailpiece receiver **23** with respect to the thumb turn **13**. Of course other arrangements are possible. For example, the tailpiece receiver **23** may be made integrally with or otherwise attached to the thumb turn **13** and need not necessarily be made separable from the thumb turn **13**. As will be understood, the tailpiece receiver **23** may be arranged to accept differently sized and/or shaped tailpieces **22** so that the door lock **1** can be used with different lock mechanisms. The tailpiece receiver **23** may be made replaceable and/or adaptable to allow for a desired range of different tailpiece or other lock mechanism engagement parts. Also, where the door lock **1** is mounted over an existing lock thumb turn, the tailpiece receiver **23** may be arranged to couple with the existing thumb turn. For example, the tailpiece receiver **23** may be arranged with a suitably sized and shaped recess to fit over the existing thumb turn and rotate the existing thumb turn based on rotation of the thumb turn **13**.

(22) While the door lock **1** in this embodiment allows a user to manually move a lock mechanism between lock and unlock positions using the thumb turn **13**, the door lock **1** may include a drive train with a motor drive function that allows the lock mechanism to be moved automatically between lock and unlock positions, e.g., in response to wireless signals from a user device such as a smartphone. For example, a lock drive may be provided to drive rotation of the tailpiece **22**, an existing thumb turn, or other lock mechanism engagement part. The lock drive may include the tailpiece receiver **23** and the drive train may be arranged to move the tailpiece receiver **23**. As an example, a drive wheel may be coupled to the tailpiece receiver **23** via the thumb turn **13**, e.g., the drive wheel may be fixed to the thumb turn **13**, and the motor may be arranged to rotate the drive wheel, and thus a coupled tailpiece **22** and lock mechanism, between lock and unlock positions. In some cases, the drive train may include a motor and transmission (e.g., one or more gears driven by the motor and engaged with the drive wheel) to move the lock drive. Control of the drive train may be performed by any suitable control circuitry, which may include a programmed general purpose

computer and/or other data processing device along with suitable software or other operating instructions, one or more memories (including non-transient storage media that may store software and/or other operating instructions), a power supply **12** for the control circuitry and/or other system components, position and other sensors, wireless communication devices to allow the control circuitry to receive and send signals with respect to user devices such as a smartphone or wireless router, input/output interfaces (e.g., such as the user interface to display information to a user and/or receive input from a user), communication buses or other links between components of the door lock **1**, a display, switches, relays, triacs, motors, mechanical linkages and/or actuators, or other components necessary to perform desired input/output or other functions.

(23) In accordance with an aspect of the invention, the door lock may include a latch that is manually movable, e.g., without tools, to attach the door lock body to the mounting bracket. For example, FIG. **4** shows a rear perspective view of the door lock **1**, and illustrates a pair of latches that are pivotally mounted to opposite sides of the body **10** and adapted to selectively engage with the mounting bracket **11**. In FIG. **4**, the latches **15** are pivoted inwardly to a closed position so that catches on each of the latches **15** engage a respective tab of the mounting bracket **11** to secure the base **17** and cover plate **16** to the mounting bracket **11**. As can be seen generally in FIGS. **1** and **4**, when the latches **15** are in a closed position, the latches **15** are positioned within an outer periphery of the body **10**, e.g., when the door lock **1** is viewed from the front, the latches do not protrude outwardly from the side portions of the body **10**. When the latches **15** are pivoted outwardly to an open position as shown in FIG. **5**, the latches **15** are extended laterally from the outer periphery of the body **10** so the latches **15** disengage tabs **111** of the mounting bracket **11**, thus disengaging the body **10** from the mounting bracket **11**. To mount the door lock **1** to a door or other surface, the mounting bracket **11** may first be separated from the body **10** and attached to the door. The mounting bracket **11** may be attached to the door in a variety of different ways, such as by adhesive, fasteners, magnets, etc. In this embodiment, the mounting bracket **11** includes a lock mechanism engaging portion **112** that has a pair of openings and pins **14** used to engage a lock mechanism. The pins **14** may include a male or female threaded portion that threadedly engages with a corresponding fastener to secure the mounting bracket **11** to a lock cylinder, cover plate or other lock mechanism structure. With the mounting bracket **11** attached to the door, the latches **15** may be pivoted to the open position and the body **10** may be positioned over the mounting bracket **11** so that mounting bracket **11** is received into a corresponding recess **171** in the base **17** and the latches **15** are appropriately positioned relative to tabs **111** of the mounting bracket **11**. The latches **15** may then be pivoted inwardly to the closed position as in FIG. **4** to attach the body **10** to the mounting bracket **11**, and thus to the door. Engaging of the latches **15** with the tabs **111** captures the tabs **111** and potentially other portions of the mounting bracket **11** between the latch **15** and the body **10**. In some embodiments, the latches **15** may be moved to the closed position by a user with a thumb and fingers of one hand, which may provide a particularly easy way to attach the body **10** to a door.

(24) While FIGS. **1-4** show an embodiment in which latches **15** are pivotally mounted to the body **10**, other arrangements for latches **15** are possible. For example, a latch may be moved linearly or along other paths than moved pivotally between open and closed positions, a latch may be mounted to a mounting bracket rather than the door lock body, and/or a latch may not be mounted to either a mounting bracket or body and instead may be separate from both elements. In one illustrative embodiment, one or more latches may be slidable or movable along a linear path to engage and disengage a lock body and mounting bracket. For example, FIG. **6** shows a rear view and FIG. **7** shows a rear perspective view of a door lock **1** in which latches **15** are movable laterally, e.g., in a plane of a door or other surface to which the lock **1** is mounted, between open and closed positions. FIGS. **6** and **7** show the latches **15** in a closed position in which tabs **111** of the mounting bracket **11** are engaged by catches of the latches **15**, e.g., the tabs **111** are captured between the latches **15** and the body **10** to engage the body **10** with the mounting bracket **11**. As can be seen in the front

view of the door lock **1** in FIG. **8**, with the latches **15** in the closed position, the latches **15** are positioned within an outer periphery of the body **10**. The latches **15** are mounted to the body **10** by fasteners **172** that are fixed relative to the body **10**. The fasteners **172** are received in respective elongated slots **151** of the latches **15** so that the latches **15** are slidably movable in a lateral direction relative to the body **10**, e.g., in a plane of the surface to which the lock **1** is mounted. FIG. **9** shows a rear perspective view and FIG. **10** shows a front view of the door lock **1** with the latches **15** in an open position in which tabs **111** of the mounting bracket **11** are disengaged and the body **10** is removable from the mounting bracket **11**. In fact, FIG. **9** shows the mounting bracket **11** removed from the recess **171** of the body **17** into which the mounting bracket **11** is receivable. In this embodiment, the base **17** includes posts **173** that are received into corresponding openings of the mounting bracket **11** when the mounting bracket **11** is received into the recess **171** of the base **17**. Engagement of the posts **173** with the mounting bracket openings may help suitably position the mounting bracket **11** relative to the base **17** and/or help prevent unwanted movement of the body **10** relative to the mounting bracket **11**. FIG. **10** illustrates how the latches **15** are extended laterally from the outer periphery of the body **10** on opposite sides of the body **10** when the latches **15** are in the open position. In this embodiment, the latches **15** may be moved toward each other to move from the open position to the closed position, and this movement may be effected by a user with thumb and fingers of one hand, e.g., squeezing the latches **15** toward each other to attach the body **10** to the mounting bracket **11**.

(25) In this embodiment, a detent is provided to hold the latches **15** in the closed position relative to the body. However, such a detent is not required, and/or a detent may be provided to hold the latches **15** in the open position in addition, or alternately, to the closed position. In this embodiment, the detent includes a pin **152** on the latches **15** (see FIG. **10**) which engages with a corresponding groove **174** (see FIG. **9**) on the body **10**. The pins **152** may be made suitably flexible to releasably engage the grooves **174**, e.g., so the latches **15** are held in the closed position until a suitably large force is exerted on the latches **15** to move them toward the open position, at which point the pins **152** disengage from the grooves **174**. Other detent arrangements are possible, such as a spring-loaded ball and groove, a pawl and ratchet, etc.

(26) In some embodiments, a latch may be guided in movement relative to the body and/or mounting bracket between open and closed positions. For example, FIGS. **9** and **10** show that the latches **15** include grooves **153** (see FIG. **10**) that receive rails **175** (see FIG. **9**) of the base **17** so that the latches **15** are guided in motion relative to the body **10**. The elongated slots **151** extend in a same direction as the grooves **153** to allow for desired movement, and the slots **151** may limit the extent of travel of the latches **15** relative to the body **10**. That is, the length of the slots **151** and engagement of the slots **151** with the fasteners **172** may limit the extent to which the latches **15** are moveable relative to the body **10**. Other arrangements are possible for guiding movement of a latch, such as reversing the relative positions of the rail and groove, providing a pocket in the base **17** into which a latch **15** moves, etc.

(27) FIG. **11** shows a front right perspective view of the latches **15** and mounting bracket **11** of the FIGS. **6-10** embodiment with the body **10** removed for clarity. The fasteners **172** are normally fixed to the base **17** and so attach the latches **15** to the base **17**. In this embodiment, springs **2** are provided at each fastener **172** to bias the latches **15** rearwardly and away from the body **10**. That is, the latches **15** are mounted to the rear side of the body **10**, and the springs **2** bias the latches **15** away from the rear side of the body **10**. This spring bias may help properly position the latches **15** so that tabs **111** of the mounting bracket **11** may be caught by catches on the latches **15** so the tabs **111** are positioned between the latches **15** and the body **10**. FIG. **11** also shows openings **113** of the mounting bracket **11** that may engage with posts **173** of the base **17** when the mounting bracket **11** is engaged with the body **10**.

(28) FIG. **12** shows a front right perspective view of the latches **15** and illustrates the catches **155** used to engage corresponding tabs **111** of the mounting bracket **11**. In this embodiment, the catches

155 include a recess formed in the latch **15** that includes a ramp or sloped leading edge **155a** to aid in engagement of a corresponding tab **111**. For example, the ramp **155a** may help guide a tab **111** into the catch **155** if the latch **15** and tab **111** are not precisely aligned. FIG. **12** also shows tabs **154** provided on a lateral side of the latches **15** that may be engaged by a tool to move the latches **15** from the closed position to the open position. Use of a tool to move the latches **15** from the closed position may be helpful, e.g., because the latches **15** may be recessed or flush with respect to the base **17** and cover **16** in the closed position, making gripping of a latch **15** by hand difficult. In addition, or alternately, a detent used to hold the latch in the closed position may provide resistance to movement of the latch from the closed position, and may require use of a tool to move the latch. The tabs **154** may be accessible at a side of the door lock **1**, e.g., a screwdriver blade may be inserted behind a tab **154** and used to pry the latch **15** outwardly from the closed position.

(29) FIGS. **13-15** show front, rear and front perspective views of the mounting bracket **11** of the FIG. **6** embodiment. These figures illustrate that the mounting bracket **11** includes a plate **114** arranged in a first plane, and having a surface, e.g., the rear side of the plate **114**, that is adapted to be positioned against or otherwise adjacent to a door surface to which the lock **1** is mounted. The tabs **111** may be arranged above the first plane of the plate **114**, e.g., extending forwardly relative to the plate **114** so the tabs **111** can be caught between the latches **15** and the body **10**. The lock mechanism engagement portion **112** may be positioned below the first plane of the plate **114**, e.g., rearwardly of the plate **114** so the engagement portion **112** may be at least partially received into an opening of a door where a lock mechanism is located. The engagement portion **112** in this embodiment includes openings for the mounting pins **14** and to allow the tailpiece **22** to pass through the mounting bracket **11**, but other arrangements are possible, e.g., depending on how the lock mechanism is arranged and/or how the mounting bracket **11** is secured to a door. While in this embodiment the mounting bracket **11** includes four tabs **111**, any suitable number of tabs **111** may be used, e.g., one on each lateral side of the bracket **11**. Also, this embodiment employs two latches **15**, but a single latch **15** may be used, and in such a case, a single tab **11** may be used. In a single latch embodiment, the mounting bracket **11** may include a hook or other feature on one lateral side that engages with an opening or other feature of the body **10**. A tab may be provided on a side opposite the hook (or opening or other feature) and engaged by the single latch.

(30) While aspects of the invention have been described with reference to various illustrative embodiments, such aspects are not limited to the embodiments described. Thus, it is evident that many alternatives, modifications, and variations of the embodiments described will be apparent to those skilled in the art. Accordingly, embodiments as set forth herein are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit of aspects of the invention.

Claims

1. A door lock comprising: a mounting bracket adapted to be mounted to a first side of a door; and a body adapted to be mounted to the mounting bracket, the body including latches movable along a linear path relative to the body to (i) an open position beyond an outer periphery of the body that permits positioning of the body adjacent to the mounting bracket and (ii) a closed position within the outer periphery of the body so that the latches engage the mounting bracket to secure the body onto the mounting bracket.
2. The door lock of claim 1, wherein the latches are movable relative to the body in a plane of a door surface to which the mounting bracket is mounted.
3. The door lock of claim 2, wherein the latches are movable relative to the body to capture a portion of the mounting bracket between the latch and the body.
4. The door lock of claim 1, further comprising a detent to hold the latches in the closed position relative to the body.

5. The door lock of claim 1, wherein the mounting bracket includes a plate arranged in a first plane, and a tab attached to the plate and positioned above the first plane, each of the latches having a catch to engage the tab to attach the body to the mounting bracket.
 6. The door lock of claim 5, wherein the mounting bracket includes a lock mechanism engaging portion that is positioned below the first plane and adapted to secure the mounting bracket to a door to which the door lock is mounted.
 7. The door lock of claim 1, wherein the latches are slidably mounted to the body, and the body has a rail or groove to guide movement of the latches relative to the body.
 8. The door lock of claim 7, wherein the body has a rail to guide movement of the latches, and each of the latches has a groove to receive and engage with the rail.
 9. The door lock of claim 8, wherein each of the latches has an elongated slot to receive a fastener to attach the latch to the body, the elongated slot extending in a same direction as the groove.
 10. A device comprising: a mounting bracket adapted to be mounted to a first side of a door; and a body adapted to be mounted to the mounting bracket, the body including latches movable along a linear path relative to the body to (i) an open position beyond an outer periphery of the body that permits positioning of the body adjacent to the mounting bracket and (ii) a closed position within the outer periphery of the body so that the latches engage the mounting bracket to secure the body onto the mounting bracket.
 11. The device of claim 10, wherein the latches are movable relative to the body in a plane of a door surface to which the mounting bracket is mounted.
 12. The device of claim 11, wherein the latches are movable relative to the body to capture a portion of the mounting bracket between the latch and the body.
 13. The device of claim 10, further comprising a detent to hold the latches in the closed position relative to the body.
 14. The device of claim 10, wherein the mounting bracket includes a plate arranged in a first plane, and a tab attached to the plate and positioned above the first plane, each of the latches having a catch to engage the tab to attach the body to the mounting bracket.
 15. The device of claim 10, wherein the latches are slidably mounted to the body, and the body has a rail or groove to guide movement of the latches relative to the body.
 16. A device comprising: a bracket adapted to be mounting to a first side of a door; and a body adapted to be mounted to the bracket, the body including a means for latching the body to the bracket, the means being movable along a linear path relative to the body to (i) an open position beyond an outer periphery of the body to permit positioning of the body adjacent to the bracket and (ii) a closed position within the outer periphery of the body to secure the body to the bracket.
 17. The device of claim 16, wherein the means for latching are movable relative to the body in a plane of a door surface to which the bracket is mounted.
 18. The device of claim 17, wherein the means for latching are movable relative to the body to capture a portion of the bracket between the means for latching and the body.
 19. The device of claim 16, further comprising a detent to hold the means for latching in the closed position relative to the body.
 20. The device of claim 16, wherein the bracket includes a plate arranged in a first plane, and a tab attached to the plate and positioned above the first plane, the means for latching having a catch to engage the tab to attach the body to the bracket.
-